

Correspondence

Statistical approximation is not general intelligence

In a Comment, Chen *et al.* argue that success in behavioural tests – including variants of the Turing test – is evidence of artificial general intelligence (AGI; see E. K. Chen *et al. Nature* **650**, 36–40; 2026). We find this problematic, on three grounds.

First, it relies on a redefinition of AGI, shifting from earlier criteria centred on robustness, generalization and reliability.

Second, it conflates performance on specific, often gameable benchmarks with a general ability to handle novelty. Benchmark success is not real-world readiness.

Third, it ignores the fact that similar outputs can arise from different processes. Large language models (LLMs) can reproduce patterns from existing data and close variations thereof. But this does not mean that they use the same kind of cognition as humans do. Faced with uncertainty, humans often express doubt or withhold judgement, whereas LLMs often respond confidently. They replace evidence-based evaluation with plausibility-driven generation. Success on superficial tests has often obscured underlying problems.

Finally, LLMs struggle with generalization and real-world performance: productivity gains remain modest, returns on investment limited, and current systems still cannot perform most human tasks. Rumours of AGI are greatly exaggerated.

Walter Quattrocio Sapienza University of Rome, Rome, Italy.
walter.quattrocio@uniroma1.it

Valerio Capraro University of Milan–Bicocca, Milan, Italy.

Gary Marcus New York University, New York City, New York, USA.

No, funding calls are not a ‘waste of time and money’

In a Career column, Gerald Schweiger argues that one Horizon Europe funding call cost more in researcher and funder time than the total amount it awarded (see G. Schweiger *Nature* <https://doi.org/qm5c>; 2025). Schweiger suggests that such schemes are a drain on the scientific ecosystem.

This analysis might be misplaced. It omits any estimate of the benefit from funded research. And it assumes that researchers would spend their time more productively if they had not been writing or reviewing proposals.

But funders anticipate that research will benefit the economy and society. Estimating the benefit precisely is complex. As an illustration, in the United Kingdom, the return is estimated at £7 (US\$9.50) for every £1 invested (see go.nature.com/45lyc1a). So a scheme costing £5 million to award £5 million in funding would produce an estimated £30 million net benefit. The cost–benefit ratio is therefore more relevant than the cost–funding one.

The key question is whether alternative funding systems exist that would maintain the quality of funded projects and be less of a drain on the research system. It is hard to imagine one that won’t involve some element of peer review or competition.

The funding system needs improving, but determining whether a grant is a ‘drain’ remains extremely difficult.

Tom Stafford Research on Research Institute, London, UK.
t.stafford@researchonresearch.org

T.S. declares competing interests; see go.nature.com/4cu2hdj for details.

Why scientists should study ‘teensplaining’

Adolescents have long been stereotyped in literature and science as being impulsive and emotional. One aspect of their unique communication style has attracted attention in popular culture: ‘teensplaining’. Similar to mansplaining, when a man condescendingly explains things to a woman regardless of her expertise, it involves youths assuming they know more than adults do (often, parents and teachers) and teensplaining things in a disrespectful tone.

This rather dismissive, although sometimes humorous, term resonates with the usual scientific view of adolescents, which focuses on shortcomings.

By contrast, supported by the growing science of adolescent development, we argue that teensplaining reflects potential adaptive roles as teenagers mature.

However, a search for ‘teensplaining’ in scientific articles using the Web of Science platform produced no hits, whereas searching for ‘mansplaining’ produced hundreds. We wish to encourage multidisciplinary teams (including linguists, psychologists, psychiatrists, paediatricians, educationalists, neuroscientists and communication scientists) to study teensplaining and what it can teach us about the learning, thoughts, emotions, motivations, identity and language development of adolescents as they transition into adulthood.

Ivy N. Defoe University of Amsterdam, Amsterdam, Netherlands.
i.n.defoe@uva.nl

Dan Romer University of Pennsylvania, Philadelphia, USA.

Smartphones are a double-edged tool in classrooms

The widespread use of smartphones by university students in China shows a conflict between conventional teaching methods and the realities of learners who grew up using digital technologies.

National strategies promote digital education and ‘smart classrooms’ that integrate technology to improve learning, yet some institutions have adopted blanket bans for phones.

Such bans might inadvertently hinder the development of digital literacy and immersive learning experiences. Classroom phone use can stem from unengaging teaching, such as unclear objectives or passive instruction. Whereas bans could help to engage more students in face-to-face interactions, they fail to address the core issue of students not physically attending classes.

Crucially, these policies overlook smartphones’ dual nature: they are potent distractors but also powerful tools for enquiry, collaboration and real-time interaction. Banning them might infringe student autonomy and foster resentment (S. McCoy and A. Marcus-Quinn *Pediatr. Res.* **98**, 1985–1987; 2025). And the subject being studied matters – some courses could require instant access to information.

Given that blanket bans fail to increase students’ interest in class and risk alienating students, strategic integration of smartphones into lessons – paired with guidance on digital citizenship – would be better.

Xiaoming Li Anhui Medical University, Hefei, China.
psyxiaoming@126.com